COURSE SYLLABUS

# Physics

2026–27 · The Cushman School


TEACHER
Mr. Willie Avendano

EMAIL
wavendano@cushmanschool.org


ROOM
Room 503

CLASS MEETS
Monday, Period 2 (8:53–9:37) · Tuesday & Thursday, Block 2 (9:36–10:56)

EXTRA HELP
Lunch on Mondays, Wednesdays, and Fridays, or mornings by appointment. *Lunch help is drop-in. For a morning slot, email the day before.*

## Course Description

A first course in physics built around measurement, modeling, and laboratory work. Students develop a shared problem-solving protocol and apply it across kinematics, forces, energy, momentum, rotation, simple harmonic motion, waves, and circuits.

The course is anchored by a sequence of five formal laboratory reports with steadily decreasing scaffolding, ending with a student-designed investigation. The emphasis throughout is on reasoning from evidence and communicating results honestly.

## Units

### Unit 0 The Physicist's Toolkit

- Units, notation, and dimensional analysis
- The problem-solving protocol
- Right-triangle trigonometry and vectors

### Unit 1 Kinematics

- Motion and graphs
- Uniform acceleration
- Cart lab — Lab Report I
- Free fall and projectiles

### Unit 2 Forces

- Free-body diagrams
- Newton's laws
- Friction — Lab Report II

### Unit 3 Circular Motion & Gravitation

- Uniform circular motion
- Orbits and gravitation

### Unit 4 Energy

- Work and power
- Conservation of energy — Lab Report III

### Unit 5 Momentum

- Impulse and momentum
- Collisions — Lab Report IV

### Unit 6 Rotation

- Torque and equilibrium
- Rotational dynamics

### Unit 8 Waves (taught before Unit 7)

- Wave properties
- Standing waves and sound
- The Doppler effect

### Unit 7 Simple Harmonic Motion

- Springs
- Student-designed pendulum investigation — Lab Report V

**Unit 9** **Circuits**

- Charge and Ohm's law
- Series and parallel circuits
- Circuits lab

Week-by-week pacing — including which weeks carry no new content — is on the course page.

## Required Texts and Links

*Subject to change and grow.*

- Course-issued readings and problem sets
- Laboratory notebook
- Scientific calculator

## Course Materials

- **TI-84 graphing calculator (required — bring every day)**
- **3-ring binder or notebook for guided notes**
- School Laptop
- Internet Connection
- Computer Charger
- Ruler
- Pencil/pen

### Software

- Desmos
- Google Sheets

## Grades and Evaluations

*Grades will be calculated using the 4.0 scale as mentioned in the student handbook.*

| 70% | 25% | 5% |
| --- | --- | --- |
| **Summative** | **Formative** | **Diagnostics** |
| unit tests, formal laboratory reports, and the student-designed investigation | problem sets, in-class practice, laboratory activities, class discussions, and unannounced quizzes | Entrance and exit tickets, and other checks that are neither summative nor formative. |

### Description of Common Assessments

**The five formal lab reports** are the backbone of the course, and the scaffolding fades across them: the kinematics cart lab is heavily guided, the friction lab semi-guided, the energy lab your first fully independent report, then collisions, and finally a pendulum investigation you design yourself.

**Unit tests** consist of conceptual questions, free response, and calculation-based problems. No extra time will be given unless there is approved accommodation, as tests are designed to be completed within the designated time.

**Honest numbers** are a graded expectation. Report the precision your measurement actually supports, distinguish exact values from approximations, and state your uncertainty. A right answer with dishonest significant figures is not a right answer.

**Classwork (CW in Veracross)** should be completed during the allotted class time and is **graded for accuracy**. Since these assessments are formative, students will have the chance to resubmit work within an allotted time determined by the teacher.

**Homework (HW in Veracross)** assignments allow students to practice their skills outside of the school setting or prepare themselves for the following class. Homework assignments are due at the start of the class period.

Scoring on homework will be as follows:

- **4** — All components of the assignment are completed in full, addressing all requirements with a high level of detail.
- **3** — Most components of the assignment are completed, meeting most of the requirements with satisfactory detail.
- **2** — Some components of the assignment are completed, but gaps or missing elements are noticeable. Detail is lacking in some areas.
- **1** — Few components of the assignment are completed, and significant gaps or missing elements are evident. Details are minimal.
- **NTI** — No assignment is turned in and is subject to the late work policy.

## Rules and Routines

*Your success in this course will directly reflect on the amount of time and energy you put into it.*

### Rules

**Respect.** This is the number one rule in the class. Respect you, your peers, and Mr. Avendano.

**Consequences.** Most violations will undergo the following consequences:

1. **1st Offense:** Verbal warning
2. **2nd Offense:** Conference at the end of class
3. **3rd Offense:** Behavior report to administration

### Routines

1. **Class preparation.** Students are expected to come to class with materials ready, including and not limited to, having their laptops charged for class, with an adequate computer charger prepared to charge their computers if need be. I will have limited materials such as pens, pencils, calculators, and paper available. However, students who are always relying on these and do not consistently have their own materials will receive scores of 1 or 2 in responsibility (IRC).
2. **Participation.** There are many ways to participate in class and I expect students to partake in some way. This can include adding to class discussions, asking questions about class materials, working collaboratively during group work, attending extra help, etc.
3. **Absences.** Consistent attendance is strongly recommended as it facilitates the learning process. It is the student's responsibility to inform me via email or in person when they are or will be absent. Students will have two class period days to make up work from absences (Student Handbook, p. 21). Any work handed in afterwards will be subject to the "Late Work Policy."
4. **Food & Drinks.** Students may not eat or drink in the classroom. Water is the only exception. If you need to eat, you may briefly step outside when I allow permission. No visits to the vending machines are allowed.
5. **Late Work.** Students will have one week to submit any late work for partial credit. Late formative assessments can receive up to a 2.0 in the gradebook. It is not fair to give the same credit to students who turn their work in on time as to those who turn their work in late. However, if a student has an *extenuating circumstance*, I am more than willing to make accommodations. See me BEFORE the due date if this applies to you! **Late work will NOT receive credit after one week.** After the week, the highest possible grade is a 0. In this case, the student will receive a Complete/No Credit or an NTI if never turned in.
6. **Resubmissions/Test Corrections.** If a student obtains a score of 2 or lower on a written exam, they may request to complete test corrections only by filling out the necessary Resubmit Form. These are done only during extra help and in my presence. Students must arrange a time with me to complete corrections within a given deadline and will earn 1/2 credit back on their missed points. The original work must have been submitted on time to resubmit. Students who want to improve an unannounced quiz score may retake a similar quiz (once). This policy only applies to the aforementioned assignments. It does not apply to regular assignments, announced quizzes, or projects.
7. **Electronics.** Cell phones must be stored in the pouch, put away in the backpack, and on silent. Students are expected to use their laptops solely for academic purposes. A student who does not comply with this expectation is not showing respect for themselves, their classmates, or their teacher and will be in violation of the classroom rule. Along with the application of consequences, the student will receive scores of 1 or 2 in industry, courtesy, and responsibility.
8. **Restroom.** If a student needs to use the restroom, they may raise their hand and point at the door. When given permission, students may go. Due to the fact that class is on the fifth floor and the restroom is only permitted for faculty on the fifth, students are permitted to go to another floor for the restroom. The time used for the restroom must be kept to a minimum and may be subject to be tracked. Excessive use of the restroom will result in students using their passes for restroom use and/or parents and administration may be informed.

## Using AI in this class

1. **Tutor, not ghostwriter.** Use it to explain, quiz you, or debug your thinking — not to produce the thing you turn in.

2. **Disclose it.** Name the tool and what it did, right on the work. Undisclosed AI is treated as plagiarism.
3. **You must be able to explain every line.** If you cannot defend it at the board, it is not yours.
4. **Verify.** AI invents numbers, citations, and units. Check against the source; units still travel with the math.
5. **Assessments are AI-free** unless the brief says otherwise.

## Academic Integrity

Students are expected and required to be honest in the preparation of assignments, papers, projects, and all other assessments. Cheating (including plagiarism) in any form is not allowed and it hinders the learning process. If a student chooses to do so, the work will receive a 0 and will lead to further disciplinary consequences. If a student cheats off a classmate's work, both will receive the above-mentioned outcomes.

## Disclaimer

The institution and teacher reserve the right to make modifications to this information throughout the academic school year, should it be necessary.

## Acknowledgement of Receipt and Understanding

Please sign and submit as soon as possible, but **no later than midnight on Sunday, August 30, 2026**.

> **Midnight means midnight.** Anything submitted at 12:01 AM Monday or later is late. Submissions are checked Monday morning.

*I have read and fully understand the academic and behavioral expectations outlined in Mr. Avendano's Physics syllabus. I agree to abide by these guidelines or will undergo the academic and/or disciplinary consequences.*

Student Printed Name _____

Student Signature _____

Parent/Guardian Printed Name _____

Parent/Guardian Signature _____

Date _____